

Trainings ▾

General Information

with Mechanically Actuated Injector

Revised diagnostic hardware was released on K38 and K50 engines in October, 1993 (engine serial number first 33125767). This included hardware installed along with a revised fuel block designed for usage with the single cylinder diagnostic kit. The original diagnostic tool was released in March, 1993 (Part Number 3824212). The kit was updated to include four needle valves designed to work in the revised fuel block.

Engines equipped with the revised fuel blocks and related hardware reduce the time involved in isolating single cylinder problems. Studies indicated a time saving of 45 minutes when using this hardware.

Preparatory Steps

with Electronically Actuated Injector

⚠ WARNING ⚠

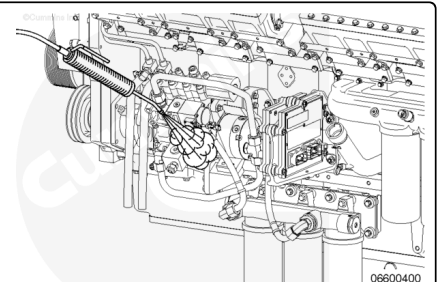
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before use.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



- Thoroughly clean debris from all fittings and components before removal. Make sure that debris, water, steam, or cleaning solution does **not** get inside the fuel system.
- Use compressed air to dry fittings and components.

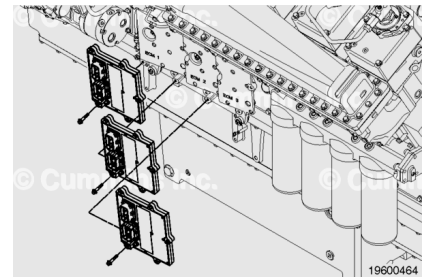
Remove

with Electronically Actuated Injector

QSK50 Engines

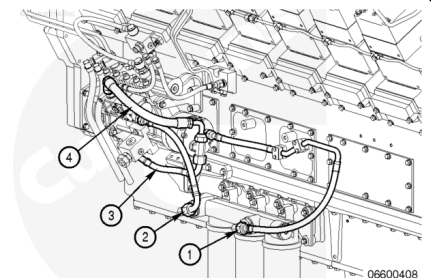
If the outlet of the gerotor to inlet of stage two filter line needs to be removed, remove the ECM. Refer to Procedure 019-031 in Section 19. (</qs3/pubsys2/xml/en/procedures/105/105-019-031.html>) This procedure is found in the QSK50 and QSK60 Modular Common Rail System, Troubleshooting and Repair Manual, Bulletin 4021533.

Remove the ECM cooling plate. Refer to Procedure 008-128 in Section 18. (</qs3/pubsys2/xml/en/procedures/28/28-008-128.html>)



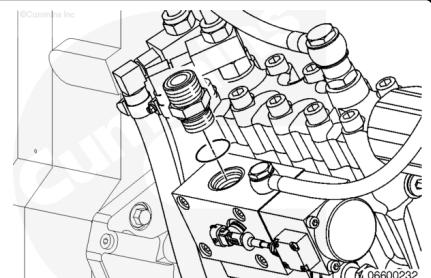
Remove the following items and cover the fittings with a plastic bag, or place the items inside a plastic bag.

1. Outlet of the gerotor to inlet of stage two filter
2. Outlet of stage two filter to the inlet of the fuel pump pressurizing assembly
3. Fuel supply to gerotor
4. Fuel bypass to gerotor.



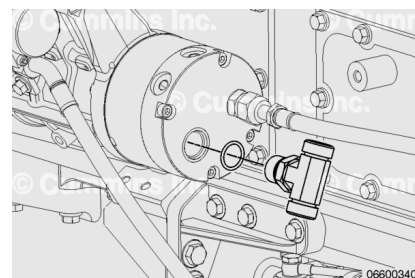
Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the union from the fuel pump pressurizing assembly and discard the o-ring.



Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

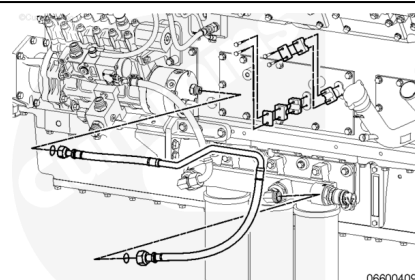
Remove the tee fitting from the gerotor fuel pump. Discard the o-rings.



Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

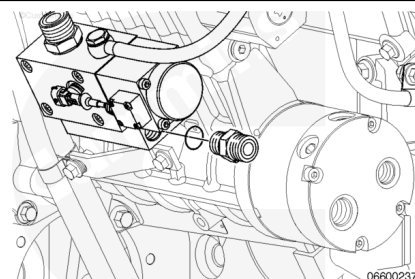
Remove the four capscrews and clamps securing the fuel line to the block.

Remove the fuel supply line from the gerotor pump output to the inlet of stage two fuel filter inlet. Discard the o-rings.

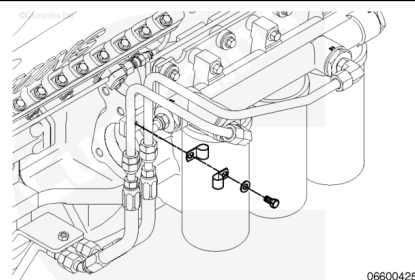


Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Remove the union from the inlet of the fuel pump actuator. Discard the o-rings.



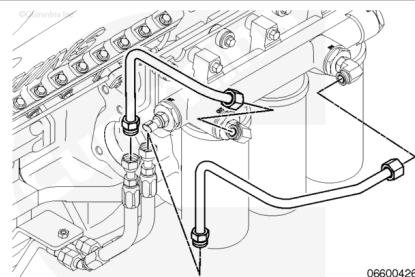
Remove the capscrews and clamps securing the fuel line to the mounting plates at the stage two filter head.



Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

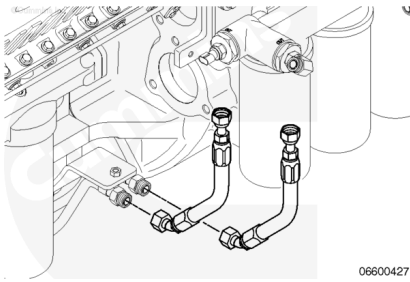
Remove the short section of the fuel supply lines to the inlet and outlet of the stage two filter head.

Discard the o-rings.

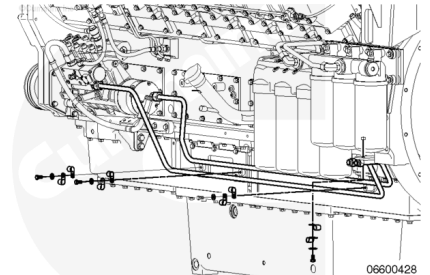


Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Remove the two flexible fuel lines.



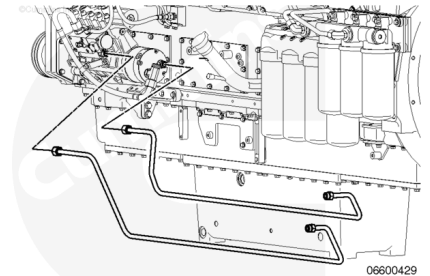
Remove the capscrews and clamps securing the fuel line to the mounting plates.



Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Disconnect the fuel supply lines at the fuel pump and remove the fuel supply lines to the remote mounted stage 2 filter.

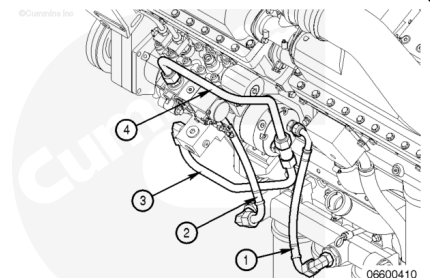
Discard the o-rings.



QSK38 Engines

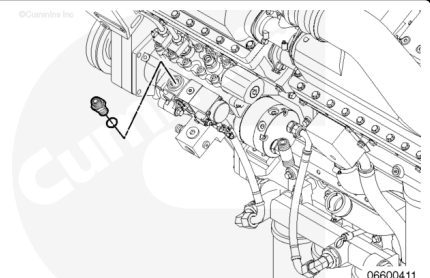
Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

1. Outlet of the gerotor to inlet of stage two filter
2. Outlet of stage two filter to the inlet of the fuel pump pressurizing assembly
3. Fuel supply to gerotor
4. Fuel bypass to gerotor.



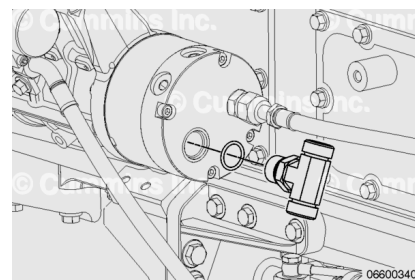
Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the union from the fuel pump pressurizing assembly and discard the o-ring.



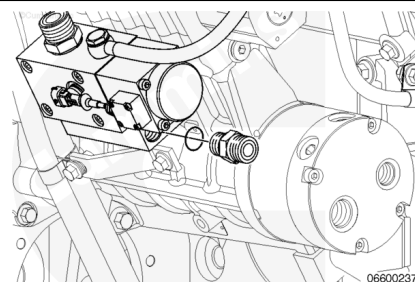
Remove the following and cover the fittings with a plastic bag, or place the items inside a plastic bag.

Remove the tee fitting from the gerotor fuel pump. Discard the o-rings.



Remove the following and cover the fittings with a plastic bag, or place the items in a plastic bag.

Remove the union from the inlet of the fuel pump actuator. Discard the o-rings.



Clean and Inspect for Reuse

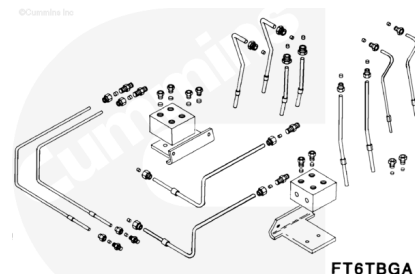
with Mechanically Actuated Injector

⚠ WARNING ⚠

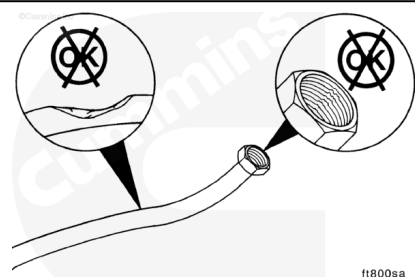
When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

Remove and discard the ferrule seals.

Use solvent to flush the tubing and the fuel block connectors.



Inspect the fuel tubes for cracks or locations where the tubing is crushed. Do **not** attempt to repair fuel tubing that is damaged.



with Electronically Actuated Injector

⚠ WARNING ⚠

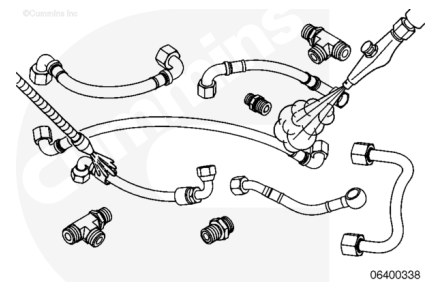
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before use.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



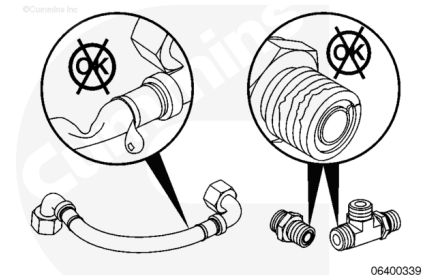
Use solvent to clean the fuel supply lines, tee fittings, and unions.

Use compressed air to dry the lines and fittings.

Inspect the fuel supply lines for restrictions that can possibly prevent fuel from flowing freely. Replace the fuel supply lines if they are kinked, or damaged, or if leakage is found.

Inspect the tee fittings and unions for damage. Replace if damaged, cracked, or stripped threads are found.

Cover the fuel line fittings with a plastic bag, or place the items inside a plastic bag after cleaning and inspection.

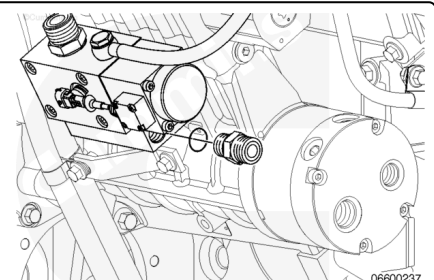


Install

with Electronically Actuated Injector

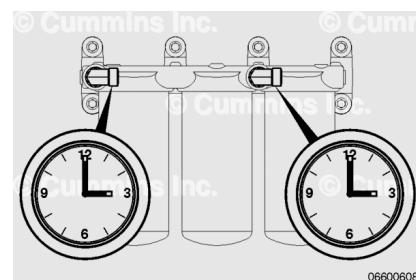
Remove the union from its protective plastic bag.

Use a new o-ring and install the union in the inlet of the fuel pump pressurizing assembly.

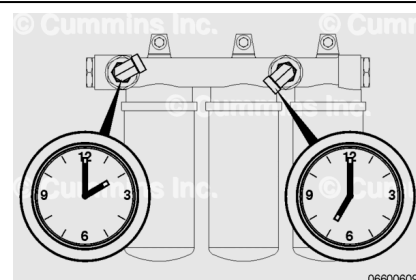


Gerotor Pump Actuator Union	100 n.m	[74 ft-lb]
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- Make sure the elbow joints are positioned at a 3 o'clock angle, as shown.
- NOTE: To reposition the elbow joints see the assembly steps in Fuel Supply Lines. Refer to Procedure 006-017 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-017.html)

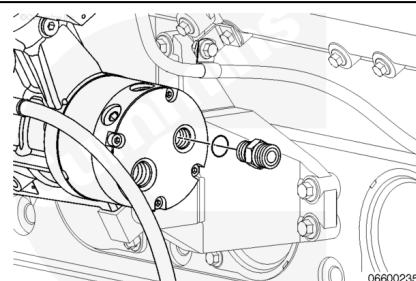


- Make sure the fuel outlet and inlet elbow joints are positioned at a 2 o'clock and 7 o'clock angles, as shown.
- NOTE: To reposition the elbow joints see the assembly steps in Fuel Supply Lines. Refer to Procedure 006-017 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-017.html)



Remove the union from its protective plastic bag.

Install new o-rings on the union and install the union in the outlet of the gerotor fuel pump.

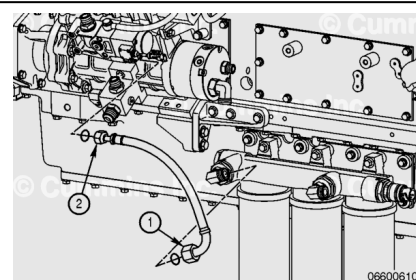


Gerotor Pump Outlet Union	100 n.m	[74 ft-lb]
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Remove the fuel line from its protective plastic bag. Install the fuel line from the outlet of the stage two filters to the inlet of the fuel pump to pressurizing assembly.

Tighten the hose fitting (1) to the elbow of the filter head.

Tighten the hose fitting at the union (2) on the fuel pump pressurizing assembly.

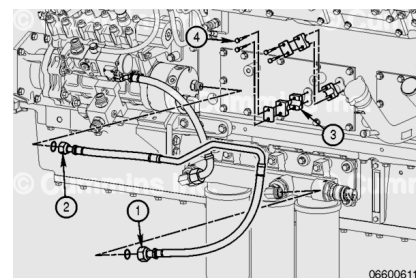


Note : Hose fitting sizes differ, depending on hose part number.

1 - 14 UNF Fittings	65 n.m	[48 ft-lb]
1 3/16 - 12 UNF Fittings	90 n.m	[66 ft-lb]

Remove the fuel line from its protective plastic bag. Install the fuel line from the gerotor fuel pump to the inlet of the stage two fuel filter.

Tighten the hose fitting (1) to the elbow of the filter head.
Tighten the hose fitting at the union (2) on the gerotor fuel pump.



Note : Hose fitting sizes differ, depending on hose part number.

1 - 14 UNF Fittings	65 n.m	[48 ft-lb]
1 3/16 - 12 UNF Fittings	90 n.m	[66 ft-lb]

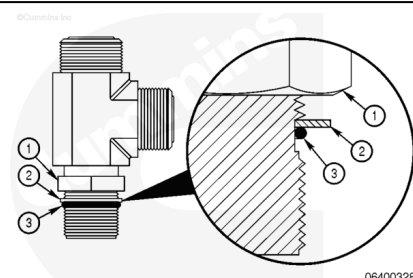
Install the clamps (3) and four capscrews (4) securing the fuel line to the cylinder block.

Fuel Supply Clamps	9 n.m	[80 in-lb]
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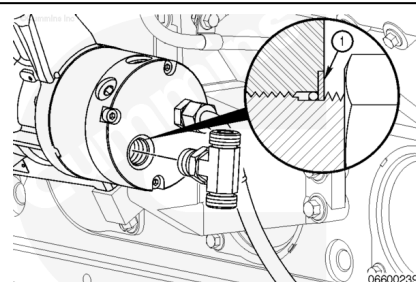
Remove the tee fitting from its protective plastic bag.

Install the tee fitting in the port on the back of the fuel pump by first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.

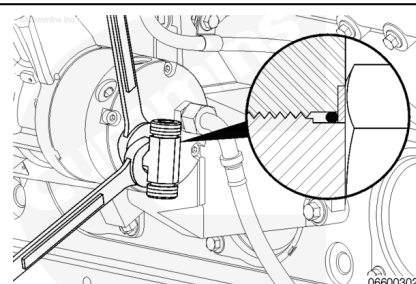


Install the tee fitting in the fuel pump port until the backup washer (1) contacts the port face.



Use two wrenches to hold the tee fitting in the desired position and tighten the locknut.

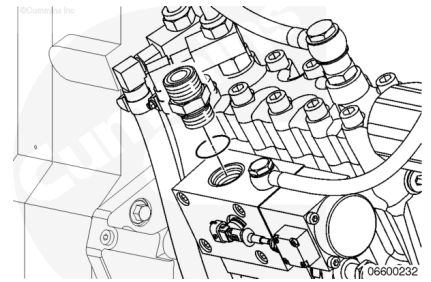
Tee Fitting Locknut	100 n.m	[74 ft-lb]
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Remove the union from its protective plastic bag.

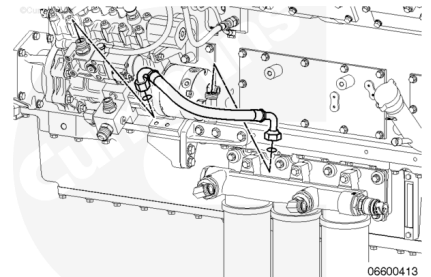
Install new o-rings in the union and install it in the port for the bypass from the fuel pump inlet actuator.

Fuel Pump Inlet to Actuator Union	170 n.m	[125 ft-lb]
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Remove the fuel supply line from its protective plastic bag.

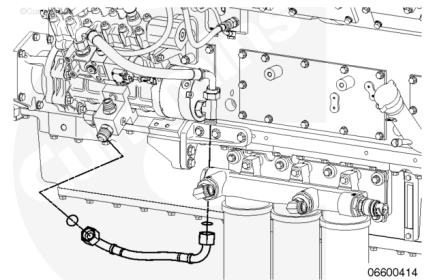
Install the fuel bypass line on the inlet actuator bypass union to the tee fitting on the inlet of the gerotor fuel pump.



Gerotor Fuel Supply Line	91 n.m	[67 ft-lb]
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Remove the fuel supply line from its protective plastic bag.

Install the fuel supply line from the fuel manifold to the tee fitting on the inlet of the gerotor fuel pump.

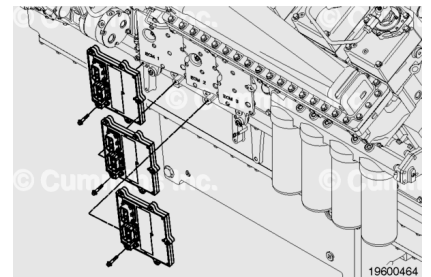


Fuel Line From Filter to Gerotor Pump Inlet	91 n.m	[67 ft-lb]
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Install the ECM cooling plate. Refer to Procedure 008-128 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-128.html>)

Install the ECM. Refer to Procedure 019-031 in Section 19. (</qs3/pubsys2/xml/en/procedures/105/105-019-031.html>)

This procedure is found in the Manual QSK50 and QSK60 Modular Common Rail System, Troubleshooting and Repair, Bulletin 4021533.

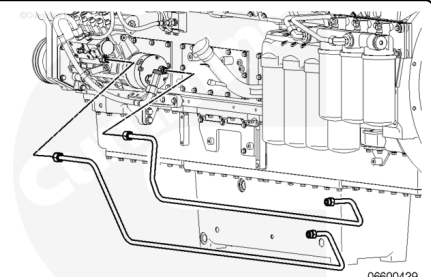


Marine Applications

Remove the fuel supply line from its protective plastic bag.

Install the fuel supply lines that go between the fuel pump and the remote mounted stage two fuel filters.

Tighten the fuel lines at the pump.

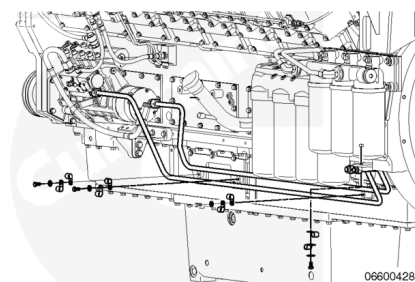


Fuel Supply Line	65 n.m	[48 ft-lb]
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Install the clamps and capscrews securing the fuel line to the mounting plates.

Tighten the capscrews.

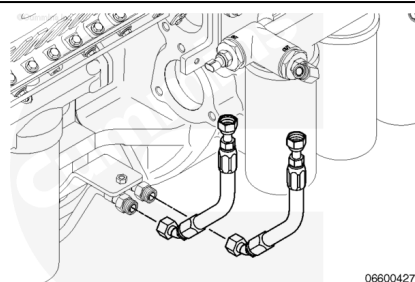
Torque Value: 9 n•m [80 in-lb]



Remove the flexible fuel supply lines from the protective plastic bag.

Install the two flexible fuel lines.

Tighten the fittings.

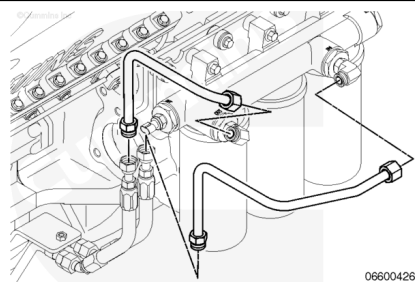


Flexible Fuel Line	91 n.m	[67 ft-lb]
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Remove the fuel supply lines from the protective plastic bag.

Install the short section of the fuel supply lines to the inlet and outlet of the stage 2 filter head.

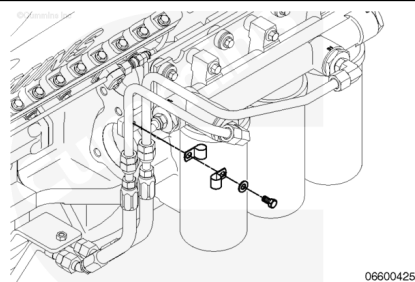
Tighten the fittings.



Fuel Supply Lines	91 n.m	[67 ft-lb]
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Install the clamps and capscrews securing the fuel line to the mounting plates at the stage 2 filter head.

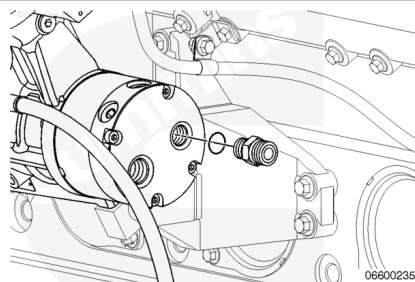
Tighten the fittings.



Stage Two Capscrews	9 n.m	[80 in-lb]
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Remove the union from its protective plastic bag.

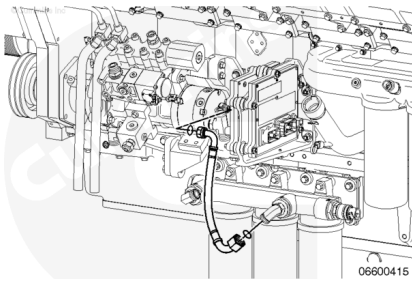
Install new o-rings on the union and install the union in the outlet of the gerotor fuel pump.



Gerotor Pump Outlet Union	100 n.m	[74 ft-lb]
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Remove the fuel line from its protective plastic bag.

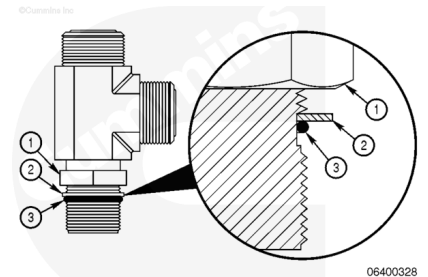
Install the fuel supply line from the gerotor fuel pump to the inlet of the second stage fuel filter.



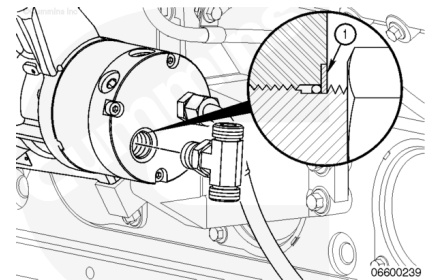
Remove the tee fitting from its protective plastic bag.

Install the tee fitting in the port on the back of the fuel pump by first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.

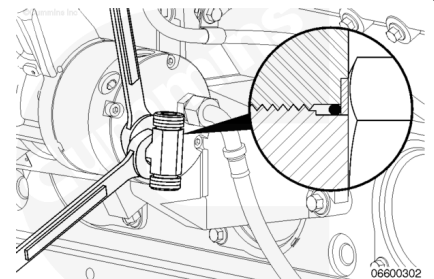


Install the tee fitting in the fuel pump port until the backup washer (1) contacts the port face.



Use two wrenches to hold the tee fitting in the desired position and tighten the locknut.

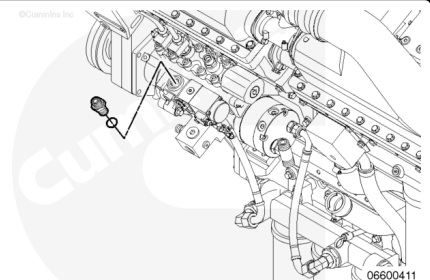
Tee Fitting	100 n.m	[74 ft-lb]
Locknut		



Remove the union from its protective plastic bag.

Install new o-rings in the union and install it in the port for the bypass from the fuel pump inlet actuator.

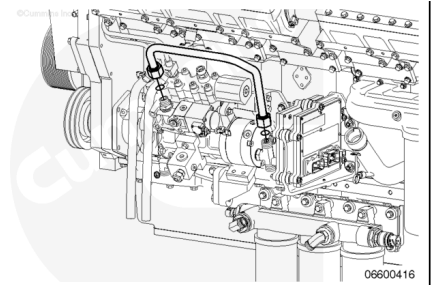
Fuel Pump Inlet to Actuator Union	170 n.m	[125 ft-lb]
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Remove the fuel supply line from its protective plastic bag.

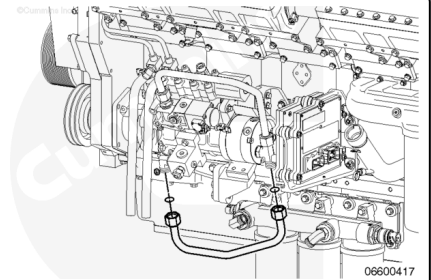
Install the fuel bypass line on the inlet actuator bypass union to the tee fitting on the inlet of the gerotor fuel pump.

Gerotor Fuel Supply Line	91 n.m	[67 ft-lb]
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Remove the fuel supply line from its protective plastic bag.

Install the fuel supply line from the fuel manifold to the tee fitting on the gerotor fuel pump.



Fuel Line From Manifold to Gerotor Pump Tee Fitting	91 n.m	[67 ft-lb]
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Finishing Steps

- Operate the engine and check for leaks. If leaks are found, repair and test again.

